

Platform server deployment solution

1. Business server resource requirements (estimate)

Metrics to collect:

1. Total quantity of goods (if the same product exists in two stores, it will be counted as two);
2. The total number of price tags, and the number of price tags of each size;
3. The frequency of price changes, the number of price changes per time, and the acceptable refresh time.
4. Number of stores; products and price tags under each store;

Some estimates are based on:

The number of goods ordered by the merchant is estimated as twice the total number of price tags. For example, if there are 100,000 price tags, the default estimated SKU is around 200,000 .

The price tag sizes are 1.54, 2.6, 4.2 in a ratio of 2:2:1, and the colors are black, white and red.

The price change frequency is calculated as 3 times a day, and each price change shall not exceed 50% of the total number of price tags.

The estimated image refresh time is within 30 minutes (from image generation to completion of image refresh).

The estimate is based on one base station with 1,000 price tags (signal strength is above -40).

If the number of stores is less than 1,000, the upper limit of price tags for a single store is estimated to be 20,000 pieces.

The server configuration recommendations are based on estimates and may be adjusted based on actual business usage.

ps: This only includes the estimate of the price tag system. If you use a middleware product to synchronize data, it is not included in this scope. The middleware needs to be configured separately.

The following configurations are for reference only and may differ from the actual values.

1K (not recommended)

Need to enable virtual memory, performance jitter will occur, not recommended

S ERVER	CPU	R AM	D ISK	Number of instances (single point)
Primary Server	4 core	8 GB physical memory 16 GB Virtual Memory	500 GB	1

ps: Using virtual memory is strongly not recommended.

The access speed of the hard disk is about 5 orders of magnitude slower than that of the memory. When the hard disk is used as virtual memory, the access speed will drop significantly and performance jitter will occur. A 16G or 32G memory stick is very low cost but very effective.

1W (not recommended)

Need to enable virtual memory, performance jitter will occur, not recommended

S ERVER	CPU	R AM	D ISK	Number of instances (single point)
Primary Server	4 core	16 GB physical memory 8 GB Virtual Memory	500 GB	1

10W

S ERVER	CPU	R AM	D ISK	Number of instances (single point)	Number of instances (cluster)
Primary Server	4 core	24 GB	500 GB	1	2

2 0W

S ERVER	CPU	R AM	D ISK	Number of instances (single point)	Number of instances (cluster)
Primary Server	8 core	32 GB	500 GB	1	2

3 0W

S ERVER	CPU	R AM	D ISK	Number of instances (single point)	Number of instances (cluster)
Application Components	8 core	16 GB	500 GB	1	2
Support components	4 core	16 GB	500 GB	1	2
database	8 core	16 GB	500 GB	1	2
cache	4 core	16 GB	500 GB	1	2

50W

S ERVER	CPU	R AM	D ISK	Number of instances (single point)	Number of instances (cluster)
Application Components	8 core	16 GB	500 GB	2	2

Support components	4 core	16 GB	500 GB	1	2
database	8 core	32 GB	500 GB	1	2
cache	4 core	32 GB	500 GB	1	2

10 0W

S ERVER	CPU	R AM	D ISK	Number of instances (single point)	Number of instances (cluster)
Application Components	8 core	16 GB	500 GB	3	3
Support components	8 core	32 GB	500 GB	1	2
database	16 core	64 GB	1 TB	1	2
cache	4 core	64 GB	500 GB	1	2

100W +

Assess resource requirements individually.

2. Monitor server resource requirements (estimates)

For scenarios with many machines, such as when the customer does not have their own operation and

maintenance monitoring system, it is recommended to deploy it to monitor server performance and identify possible performance bottlenecks.

It is recommended to deploy 300,000 or more servers. The configuration is estimated based on 1-2 servers, 8 cores + 32G + 1T hard disk.

3. LCD Server Resource Requirements (Estimated)

For the business characteristics of LCD devices, it is recommended to deploy LCD servers separately for 5k and above quantities. The configuration is calculated in multiples of 4W.

4 W

S ERVER	CPU	R AM	D ISK	Number of instances (single point)	Number of instances (cluster)
Primary Server	8 core	16 GB physical memory	500 GB	1	2

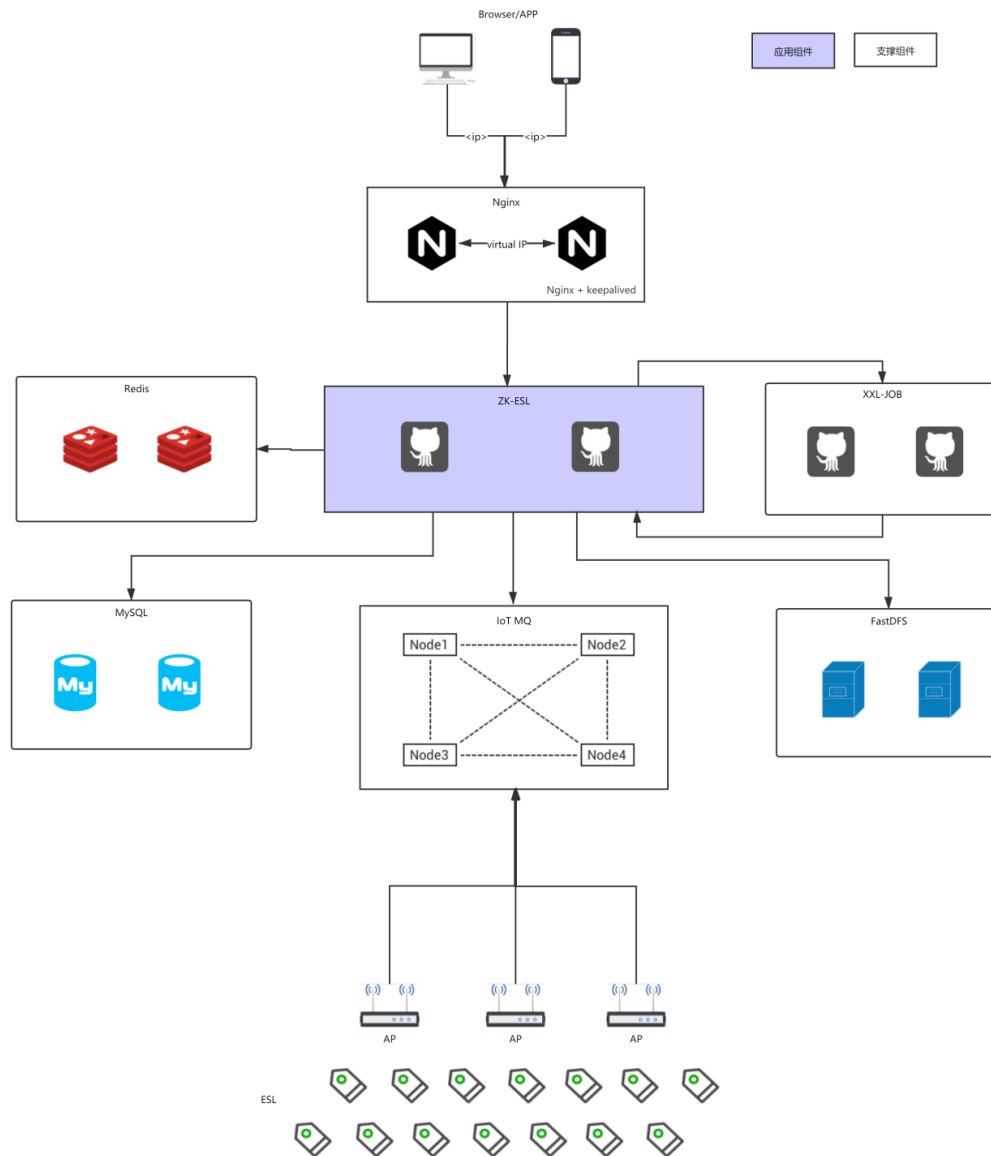
8 W

S ERVER	CPU	R AM	D ISK	Number of instances (single point)	Number of instances (cluster)
Primary Server	8 core	16 GB physical memory	500 GB	2	2

8W +

Assess resource requirements individually.

4. Deployment architecture diagram



The application component is the core of the entire system and completes all business logic. The support component provides service support for the application component.

Component List

Component Name	Component Type	use
nginx	Support components	Reverse proxy + static service container

xxl-job	Support components	Task Scheduling Center
Redis	cache	cache
MySQL	database	database
FastDFS	Support components	File storage service
emqtt	Support components	Message Component
canal	Support components	Data change monitoring service
zk-esl	Application Components	Core business logic services, complete business logic processing except LCD products
zk-lcd	Application Components	Core business logic service, complete LCD related business logic processing
gateway	Support components	Access Gateway

Versions of each component

Component Name	Component Version	Remark
OS	22. 04	Ubuntu 22. 04. 5 LTS
NGINX	1. 2 4. 0	
MySQL	8. 0. 28	
Redis	6. 2. 10	
Fastdfs	5. 11	
Canal	1. 1. 6	
EMQX	4. 4. 16	

5. Environmental requirements

Operating system: Centos7.5 or 7.6 , esl-v-6.2.0 also supports Ubuntu 20.04.6 LTS or 22.04.4 LTS

Hard Drive: SSD

CPU: Main frequency 2.6 GHz or above

Bandwidth: Intranet 1000Mbps +, public network 100Mbps+ ; store bandwidth 20M bps + ;

Disable virtual memory, disable TCP connection multiplexing in NAT environment, etc.
 selinux and firewalld are enabled , you need to shut down and restart the system before deployment.

Supports static IPv4 addresses, smooth intranet access to the server, and smooth internal access to all ports.

Cluster deployment needs to support keepalived components and be able to map the entry IP to the virtual IP provided by keepalived, while providing public network entry load balancing services. If it is a cloud server, it is recommended to provide cloud products such as MySQL (version 5.7.27 or above) and Redis (version 4.0.14 or above) . (Some cloud service providers will disable the keepalived service)

Port occupation and opening status

Serve	Occupied port number	Open ports	use
All Servers	twenty two	twenty two	SSH remote login port. It is recommended to change it to a non-default port and update the password regularly or use certificate access
nginx	80, 443	80 , 443	Front-end access port, http or https
nginx	1883, 10001, 7778 8086, 9989, 8898 18083、8411、8301	1883 , 8086, 18083	[Optional] Cluster service nginx proxy port . Non-cluster architecture does not need to open it.
nginx	10101-10110	unnecessary	Reserve 10 ports , such as monitoring service access
xxl-job	8085 , 8086	8085 , 8086	Scheduling service port
Redis	6379	unnecessary	Service Port
MySQL	3306	unnecessary	Service Port
FastDFS	8300 , 8301, 8410 , 8411	unnecessary	Service Port
emqtt	1883, 2883, 5369, 8081 , 8083, 8084, 8883 , 11883, 18083, 18084	1883 , 2883 , 8083, 18083 , 18084	Service Port
canal	11110 , 11111, 11112	unnecessary	Service Port
zk-esl	9999 , 9988, 9977	unnecessary	Service Port
zk-lcd	7777、8899	unnecessary	Service Port

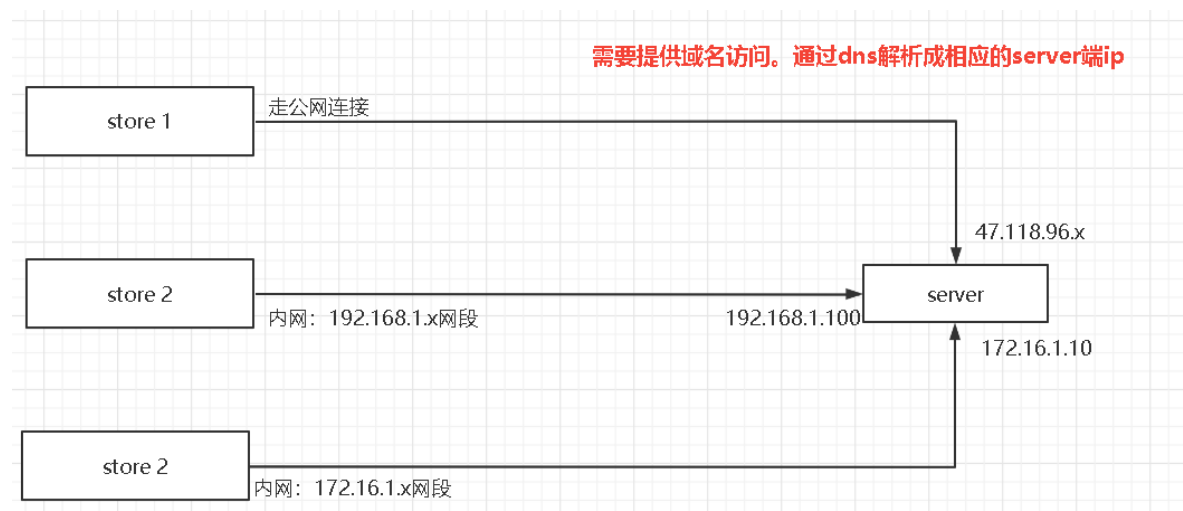
Additional notes:

1. As the version iterates, the number of internally occupied ports may increase. It is recommended that the intranet ports be interconnected without restrictions.
2. The port will be temporarily occupied during the service operation;
3. The port needs to be forwarded based on TCP;
4. Be sure to provide a normal access speed + reliable remote method (ssh) to facilitate after-sales problem handling support;
5. It is not recommended to deploy a self-built server based on a virtual machine. This will have a significant impact on performance;

6. Other additional requirements / suggestions

If there is a domain name, it is recommended to use the domain name first. System access address, AP connection address, etc.

If the store and the server communicate, it involves using multiple network segments at the same time (which can be understood as multiple network cards on the server), or using the internal and external networks at the same time. Domain name access needs to be provided and resolved into different IP addresses through DNS service.



7. Version update notes

Version	Modification time	illustrate	Revised by
v1.0	2021-12-20	Initial release	
v1.1	2022-02-09	Improve server resource demand assessment	
V1.2	2022-07-20	Improve environmental requirements	
V1.3	2022-08-04	1. Supplement monitoring server resource requirements; 2. Improve the business server resource assessment instructions	Liu Xin
V1.4	2022-11-18	1. Supplementary environmental requirements 2. Supplementary description of internally used ports 3. Added "Other Supplementary Requirements" section	Liu Xin
V1.5	2023-03-27	1. Update machine resource configuration 2. Resource allocation evaluation to increase the number of price tags of 1k and 1w 3. Increased store bandwidth requirements	Liu Xin
V 1.6	2023-05-11	1. Update the version description of each component	Song Yifei
V1.7	2023-12-20	1. Update some component versions 2. Add new ports and supplement cluster ports	Xu Xu
V1.8	2024-05-14	1. Updated operating system requirements	Li Jun
V1.9	2024-12-24	1. Update 50w resource configuration 2. Update the version information of each component	Song Yifei
V2.0	2025-03-20	1. Adjust the database server configuration 2. Added LCD server requirements	Xu Xu